

Calibration of Low-Cost Particle Sensors by Using Machine-Learning Method

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Abstract—The measurement of particle matter (PM) of mass concentration by low-cost PM sensor is strongly influenced by environmental factors such as humidity, temperature, wind speed, wind direction. In this study, we developed a machine learning-based calibration method for low-cost light-scattering PM sensor. A Feedforward Neural Network (FNN) was used to compensate for the effect of environmental factors on the PM measurements. Experimental data were collected from 20 March – 6 May 2018 in central Taiwan, and used to train and evaluate the calibration model. Before calibrating PM sensor, the PM_{2.5} mass concentration of low-cost PM sensors have the lowest values of R-squared (R^2), with 0.618 ± 0.033 as compared to the Environmental Protection Agency (EPA) approved federal equivalent method (FEM) instrument (BAM-1020, Met One Instruments). After calibrating PM sensor by using the FNN calibration model, the PM_{2.5} mass concentration of low-cost PM sensors show the highest linearity with an R^2 value of 0.905 ± 0.013 for BAM-1020. It demonstrated that the machine-learning method could be used to calibrate a low-cost PM sensor and improve its accuracy.

Keywords—Particulate matter, low-cost sensor, calibration, machine learning, artificial neural network

I. INTRODUCTION

Particulate matter (PM) air pollution can have a significant impact on human health, exposure to particulate matter over long periods can cause asthma, respiratory inflammation, jeopardizes lung functions and even promotes cancers.^[1] Therefore, most countries in the world strictly and continuously monitor ambient airborne particulates to protect human health. Particulate matter can be classified as PM₁₀, PM_{2.5}, or PM₁, according to the aerodynamic sizes of less than 10 μm , 2.5 μm , and 1 μm , respectively.^[2] The national ambient air quality standards (NAAQS) for the PM are specified based on the mass concentration of PM_{2.5} and measured by either a federal reference method (FRM) or a federal equivalent method (FEM).^[3] For example, Beta Attenuation Mass Monitor (BAM-1020, Met One Instruments) and tapered element oscillating microbalance (TEOM® 1405-F, Thermo Scientific) are the U.S. Environmental Protection Agency (EPA) FEM monitoring system, these instruments can automatically measure and record particulate matter mass concentration. However, these devices cost tens of thousands of dollars and require specific infrastructure and trained people to operate.^[4] Within the last ten years, more and more low-cost PM sensors are used to measure particulate matter mass concentration. Although low-cost PM sensors

cannot replace expensive FRM/FEM instruments, these sensors have created new opportunities for providing high spatial and temporal resolution PM information and personal health.^[5]

However, most low-cost PM sensors have not been thoroughly evaluated with standardized calibration protocols. To improve accuracy in the measurement of PM mass concentration by low-cost PM sensor, it is necessary to establish quantitative calibration relationships against FRM/FEM approved reference instrument. This is usually done under controlled laboratory environment or co-location alongside a reference instrument in the field.^[2,6] However, most low-cost PM sensors were weakly correlated ($R^2 < 0.3$) with a tapered element oscillating microbalance (TEOM)^[7] because low-cost optical PM sensors are very sensitive to environmental factors such as humidity, temperature, wind speed, wind direction in the field.

In this study, we aim to improve the calibration strategies of low-cost PM sensors using three calibration models (Linear Regression, Support Vector Machine (SVM), and Feedforward Neural Network), which, to our knowledge, has not been previously applied to low-cost PM sensors calibrations. To ensure calibration model robustness, we were developed and validated four low-cost PM sensors against FEM approved reference instrument. Furthermore, the study was conducted over a 60-day period spanning multiple seasons and a wide range of meteorological conditions in the field.

II. EXPERIMENTALS

Experiments were carried out around the Tunghai University, in Taichung City (24.181838, 120.595857). Our system and stations provided by the Environmental Protection Bureau collect data at the same time, as shown in figure 1. The station is equipped with meteorological sensors (temperature, relative humidity, wind speed and wind direction) and reference PM instruments (BAM-1020, Met One Instruments) for PM 2.5 and PM 10. These reference measurements were used for low-cost PM sensor calibration and data validation.

(A) System Architecture

The proposed PM data logger consists of four low-cost PM sensor nodes, an LTE gateway (MOXA G3470A), a wireless access point (MOXA AWK3131), and a UDOO (UDOO QUAD) computer system. First, the UDOO system and four low-cost PM sensor nodes are actively connecting

to the access point and then the UDOO system inspects each low-cost PM sensor nodes connection status. When TCP connection is created, the UDOO system sends the read command (0XA3 0X00) to each PM sensor node and then waits for response data of each PM sensor node. The sending duration time of each PM sensor node is 5 minutes for one time which is controlling by the PM sensor node. The UDOO system transmits the data to the remote server by LTE gateway

The low-cost PM sensor node (as shown in Figure 1) includes a microprocessor, a PM2.5 sensor (PMS7003, Plantower) and a digital Relative Humidity sensor with a Temperature output (HTU21D, Measurement Specialties), it also equipped with wireless modules for data transmission. In the sensor node architecture, the microprocessor is in charge of the sensor data collecting and transmitted to the wireless module. The PM2.5 sensor is a low-cost light-scattering sensor with UART interface, which can detect particles of different diameter from 0.3 μm to 10 μm .



Figure 1. photographs of the custom-designed PM data logger (left) and low-cost PM sensor node (right).

(B) Prediction Methods

Linear Regression

Linear regression is a simple and important method for predictions in the analysis. The simplest form of an equation for a straight line could be expressed as below:

$$y = ax + b \quad (1)$$

Where a is the slope and b is the y-intercept. When we want to fit a linear equation to certain data with multiple variables as our case – PM mass concentration, temperature, humidity, etc., the equation for the straight line could be expressed as below:

$$y = a_0x_0 + a_1x_1 + a_2x_2 + \dots + a_nx_n + b \quad (2)$$

Where n is the number of the variables. The Linear Regression model provided in scikit-learn offers an easy way to retrieve the corresponding a_i where $i = 0, 1, \dots, n$, and b . The calculation needed for linear regression is low, the method is great for the cases with high linearity. In low linearity cases, you can imagine it is hard to find optimized a_i for the equation, thus the output and the target could get great differences.

Support Vector Machine

Support Vector Machines (SVMs) are a popular method in machine learning. They can be used for classification,

regression and distribute estimation. Assume we have a large portion of observation data, and the values of observed PM mass concentration are distributed from 0 to 200, the estimation model can be achieved by simply classifying the input data into 0 to 200 since the observed PM mass concentration values are always positive integers.

Given a training set of observing data and PM mass concentration pairs $x_i, y_i, i = 1, 2, \dots, l$ where $x_i \in R^n$ and $y \in \{1, -1\}^l$, the SVM solves the following optimization problem:^[8]

$$\min_{w,b,\xi} \frac{1}{2} w^T w + C \sum_{i=1}^l \xi_i \quad (3)$$

Subject to $y_i(w^T \phi(x_i) + b) \geq 1 - \xi_i$ where $\xi_i \geq 0$

The function ϕ maps the training vector x_i into a higher (maybe finite) dimensional space, the SVM finds a linear hyperplane with the maximum separating margin in this higher dimensional space. $C > 0$ is the penalty parameter of the error term. In our case, the problem may be solved by linear approach, thus the kernel function can be used by the linear one as the following:

$$K(x_i, x_j) \equiv \phi(x_i)^T \phi(x_j) = x_i^T x_j \quad (4)$$

Feedforward Neural Network (FNN)

If we got a large amount of observation data, using a neural network for predicting the after coming data is a feasible idea. In our case, the neural network model consists of two hidden layers, each hidden layer contains 512 neurons, as the Figure 2 shown below:

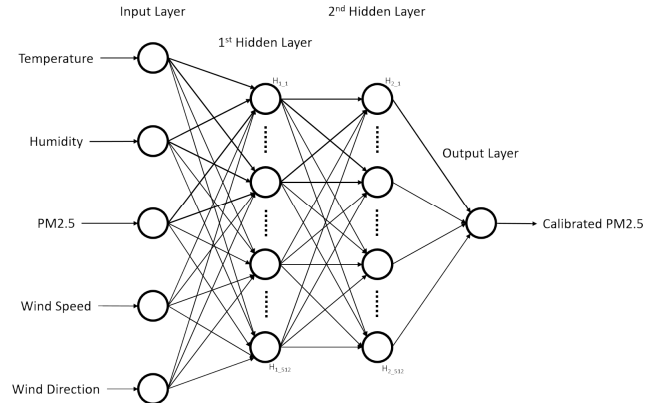


Figure 2. An illustration of a Feedforward Neural Network with two hidden layers.

Given a training set of observing data and PM mass concentration pairs $x_i, y_i, i = 1, 2, \dots, l$, the requirement is to calculate for the parameters of the neural network so the predicted output $\hat{y}_i$ is as close as possible to the actual output y_i , statistically. The performance can be measured by the mean squared error:

$$MSE = \frac{1}{l} \sum_{i=1}^l (\hat{y}_i - y_i)^2 \quad (5)$$

All the layers use Keras 'normal' as the kernel initializers, 'relu' as the activation functions, and 'adam' as the optimizers. PM2.5 values are rounded to integers after

interpolation, all other values use single-precision floating-points for computations.

(C) Training Dataset

The raw data selected from our sensors are from the date 2018/03/20 to 2018/05/06, where the sensors measured a sample every 5 minutes. The hourly average data (ground-truth) is obtained from the Environmental Protection Bureau (EPB). In order to increase the size of the dataset, we have interpolated the data from the EPB using cubic spline data interpolator in SciPy (SciPy.org) to 12 samples an hour, namely, there is an interpolated sample every minute after the interpolation. Finally, we got 13,809 samples of training data in total (invalid samples are eliminated).

III. RESULTS AND DISCUSSION

Some of the raw data (PM mass concentration, humidity, and temperature) from for sensor node are given in Figure 3. A data series from this sensor node was a span from 20 March – 6 May 2018. Daily temperature/relative humidity cycle was easily observed in Figure 3. We also know that the optical PM sensors are easily sensitive to humidity and relative humidity, so we will calibrate them later. Figures 4 presents graphically and statistically the pairwise correlations between all the low-cost sensors' PM2.5 mass concentrations. The PMS7003 sensors were well correlated with each other (R^2 ranging from 0.86 to 0.97).

The training dataset is split into two datasets – one with 12,000 samples for training model, the rest of it for verification. The training data is shuffled before feeding into the training model, to eliminate the effects of the particular data order information from being learned by the neural network model. In the first run, we only put temperature, humidity, and PM mass concentration as input parameters. In the second run, we add another input parameters (wind speed and wind direction) for training model. The scatter plots of the original (uncalibrated), the calibrated (via Linear Regression, SVM, and FNN) are shown in figure 5.

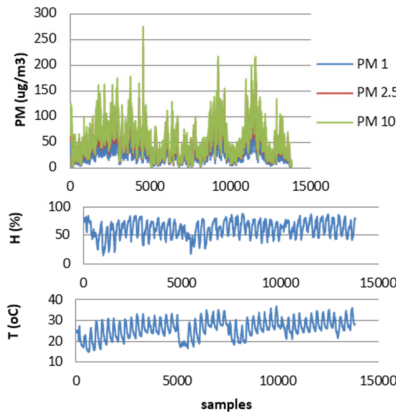


Figure 3 Time series plots of (top) PM concentration and (middle) humidity and (bottom) temperature with a five-minute resolution in the field (2018/3/20 to 2018/5/6).

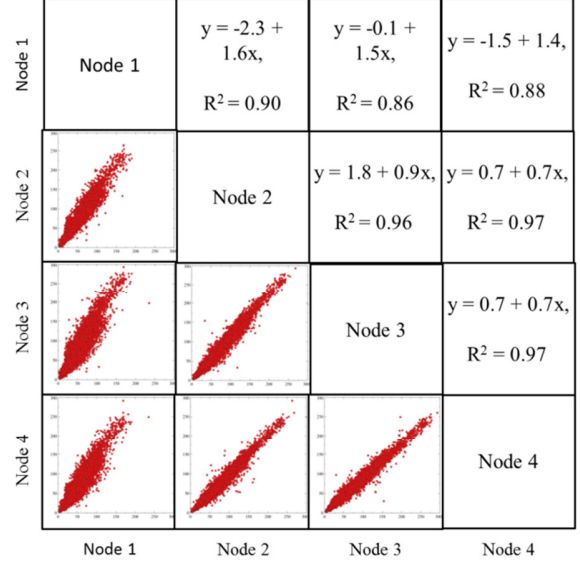


Figure 4. Pairwise correlation between PM2.5 mass concentration ($\mu\text{g}/\text{m}^3$) of uncalibrated low-cost PM sensors. Upper-right section: fitting equation and R^2 of linear regression models using the ordinary least squares (OLS) method. Lower-left section: linear regression lines superimposed on pairwise plots.

The linear regression model only solves the multivariate linear equation for fitting the data, for the far-from-linear and nasty input (such as the PM10 data we used here), the lower the accuracy of the calibration is quite expectable, though it is the fastest method against the others. Due to the features of the SVMs that SVMs do not directly provide probability estimates, so SVMs only classify the inputs into the output already known by the model. In our case, the SVM classifies our testing dataset into our training dataset (known), if the testing dataset contains some radical or completely different kind of dataset, the SVM model would still classify them into the already known ones, which lowers the accuracy rate. From the results, we can say that the two-hidden-layered feedforward neural network we proposed on Keras is quite effective on data calibration – especially when the input data is more sufficient and correlated. The reason we chose to use 512 neurons in FNN is due to performance-computation trade-off. In our experiments, the FNN model with more neurons generally produces better results as well as more time-consumption.

Using the R-squared (R^2) of the linear regression model as indicators of model fit, as shown in Table 1. The feedforward neural network model that included relative humidity (RH %), temperature ($^{\circ}\text{C}$), PM mass concentration ($\mu\text{g}/\text{m}^3$), wind speed and wind direction was found to best predict uncalibrated raw PM mass concentration to into BAM-1020 PM mass concentrations.

Table 1. Calibration models comparison using R^2 of the linear regression model

Input data	Model	R-squared (R^2)	
		PM 10	PM 2.5
PM	w/o calibration	0.234 ± 0.032	0.618 ± 0.033
Temperature Humidity PM	Linear Regression	0.373 ± 0.037	0.701 ± 0.039
	SVM	0.486 ± 0.051	0.762 ± 0.043
	FNN	0.705 ± 0.034	0.839 ± 0.036
Temperature Humidity PM Wind Speed Wind Direction	Linear Regression	0.467 ± 0.020	0.728 ± 0.030
	SVM	0.756 ± 0.032	0.857 ± 0.019
	FNN	0.882 ± 0.029	0.905 ± 0.012

IV. CONCLUSIONS

This study demonstrated that the two hidden layers of feedforward neural network model could be used to calibrate the low-cost PM sensors. The accuracy of the low-cost PM sensors can be improved with further on-field calibration and validation. Thus, future work will collect more data over one year in the field, and prepare datasets obtained from a range of environments. We will further focus on the study of other data type (such as Sox, NOx, O3, etc.,) effects on the accuracy of the calibrated low-cost PM sensor by machine learning method.

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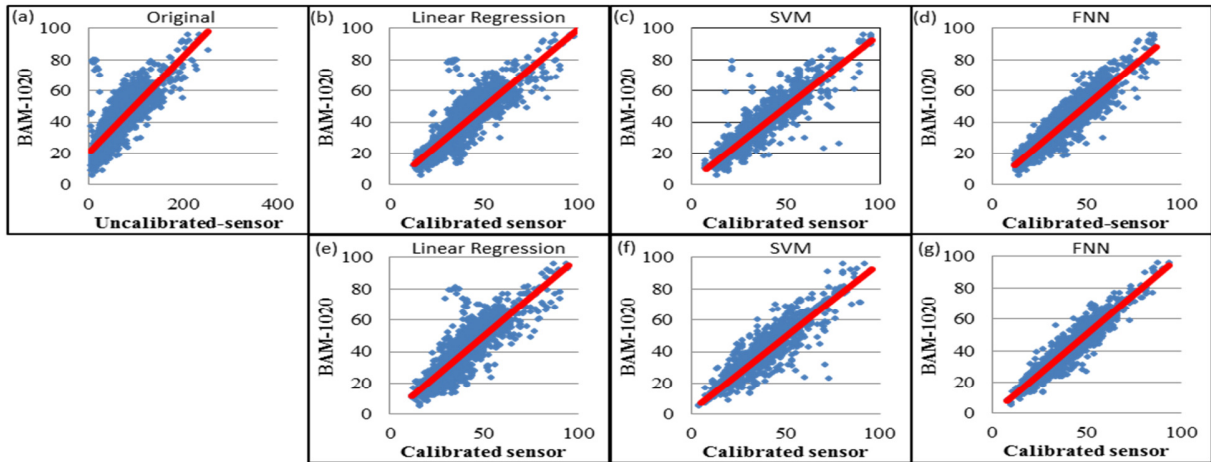


Figure 5. The scatter plots of the (a) original PM sensor, and the calibrated PM sensor (via (b)Linear Regression, (c) SVM (d) FNN) against BAM-1020.